

Muhamad Assem Ahmed

Lab 16

Part A

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Student Portal

Every row below travelled from SQL Server, through Entity Framework, into a controller that never called `new`.

Today, the same rows, reachable at `/students` — an address we design ourselves.

Full Name	Year	GPA
Kareem Fouad	4	3.20
Nada Samir	1	3.60
Omar Hesham	3	2.80
Yara Adel	2	3.50

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All Students

Reached through the route pattern `students`.

Id	Full Name	Year	GPA
4	Kareem Fouad	4	3.20
3	Nada Samir	1	3.60
2	Omar Hesham	3	2.80
1	Yara Adel	2	3.50

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Nada Samir

Reached through the route pattern `students/{id:int}`.

Id	3
Full Name	Nada Samir
Year of Study	1
GPA	3.60

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Year 2 Students

Reached through the route pattern `students/year/{year:int:range(1,4)}`.

Full Name	Year	GPA
Yara Adel	2	3.50

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Band: first

Reached through the route pattern `students/honours/{band:honourband}`.

Full Name	Year	GPA
Nada Samir	1	3.60
Yara Adel	2	3.50

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The `default` route sits at the bottom because conventional routes are matched top-to-bottom, first match wins

Part B

All Students

Reached through the route pattern `students`.

Id	Full Name	Year	GPA
4	Kareem Fouad	4	3.20
3	Nada Samir	1	3.60
2	Omar Hesham	3	2.80
1	Yara Adel	2	3.50

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All Students

Reached through the route pattern `students`.

Id	Full Name	Year	GPA
4	Kareem Fouad	4	3.20
3	Nada Samir	1	3.60
2	Omar Hesham	3	2.80
1	Yara Adel	2	3.50

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It's acceptable for two URLs to reach the same action as long as each has a clear purpose

Part C

All Students

Reached through the route pattern `students`.

Id	Full Name	Year	GPA
3	Nada Samir	1	3.60
1	Yara Adel	2	3.50

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```

[START] Request path : /students/top/2
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (4ms) [Parameters=[@p='2'], CommandType='Text', CommandTimeout='30']
      SELECT TOP(@p) [s].[Id], [s].[FullName], [s].[Gpa], [s].[YearOfStudy]
      FROM [Students] AS [s]
      ORDER BY [s].[Gpa] DESC
info: 8/2/2026 14:13:35.312 RelationalEventId.CommandExecuted[20101] (Microsoft.EntityFrameworkCore.Database.Command)
      Executed DbCommand (4ms) [Parameters=[@p='2'], CommandType='Text', CommandTimeout='30']
      SELECT TOP(@p) [s].[Id], [s].[FullName], [s].[Gpa], [s].[YearOfStudy]
      FROM [Students] AS [s]
      ORDER BY [s].[Gpa] DESC
[END] Request path : /students/top/2
[START] Request path : /lab_16.styles.css
[END] Request path : /lab_16.styles.css
[START] Request path : /students/top/5
[END] Request path : /students/top/5
[START] Request path : /students/top/acc
[END] Request path : /students/top/acc

```

MAX_YEAR (4) IS accepted range(1,4) is inclusive on both ends, so /students/top/4 matches;
/students/top/5 fails the constraint and 404s before the action

Part D

← → ↻ 🔍 <https://localhost:7019/students/intake/tiaA> ☆ ... 🔍 Search

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All Students

Reached through the route pattern **students**.

Id	Full Name	Year	GPA
1	Yara Adel	2	3.50
2	Omar Hesham	3	2.80
3	Nada Samir	1	3.60
4	Kareem Fouad	4	3.20

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← → ↻ 🔍 <https://localhost:7019/students/intake/tiaB> ☆ ... 🔍 Search

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Part E

← → ↻ 🔍 <https://localhost:7019/Students/About> ☆ ... 🔍 Search

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← → ↻ 🔍 <https://localhost:7019/about/muhammed> ☆ ... 🔍 Search

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All Students

Reached through the route pattern **students**.

Id	Full Name	Year	GPA
4	Kareem Fouad	4	3.20
3	Nada Samir	1	3.60
2	Omar Hesham	3	2.80
1	Yara Adel	2	3.50

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All Students

Reached through the route pattern `students`.

Id	Full Name	Year	GPA
3	Nada Samir	1	3.60
1	Yara Adel	2	3.50

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/ Students / About is a 404 because attribute routing replaces conventional routing

minGpa belongs in the query string rather than the path because it's optional filter on the resource, not part of identifying the resource itself

Part F

Lab ID 27

$$\text{MAX_YEAR} = (27 \bmod 4) + 1 = 3 + 1 = 4$$

$$\text{MIN_GPA} = 2.5 + (27 \bmod 3) \times 0.5 = 2.5 + 0 = 2.5$$

$$\text{INTAKE_CODE} = 27 \bmod 3 = 0 \rightarrow \text{itiA}$$

Requesting `/students/top/5`: the request hits the routing middleware, which checks the pattern `students/top/{count:int:range(1, 4)}`. The int constraint parses "5" successfully, but the range constraint rejects it because $5 > 4$. Since no route matches, the request never reaches the controller/action

Same: both the built-in `int`/'range' constraints and my `IntakeCodeConstraint` implement `IRouteConstraint` and are evaluated by the routing middleware in the same way before any action executes

It's a guarantee, not a limitation: once an action has its own `[Route]`